

JULIEN GLORY MANANA

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PROFILE

Python backend developer and Computer Science student building AI-powered applications with FastAPI, Django, PostgreSQL, and LLM APIs. Experienced in designing REST APIs, integrating OpenAI/Claude/Azure OpenAI, building RAG pipelines, and shipping deployed full-stack products across fintech, healthtech, and education. Seeking backend, software engineering, or AI engineering internship opportunities.

EDUCATION

BSc Computer Science

Expected Sept 2028

Kwame Nkrumah University of Science and Technology (KNUST) · Kumasi, Ghana

- Mastercard Foundation Scholar
- Relevant coursework: OOP (Java), C++, SQL, Operating Systems, Database Systems, System Analysis & Design

W.A.S.S.C.E (Sciences)

Sep 2020 – Jun 2023

Rochas Foundation College of Africa · Nigeria

EXPERIENCE

AI & Software Engineering Trainee

Feb 2026 – Present

Next Generation of Tech Giants (nGOT Giants) · Kumasi, Ghana

- Completed a 10-day AI/ML and Agentic Systems intensive at KNUST's College of Engineering Innovation Centre — shipping 20+ projects across fintech, healthtech, agritech, and ecommerce at 12 hrs/day.
- Building backend for Port, an internal collaboration tool, within a cross-functional engineering team.

Intern

Nov 2025 – Dec 2025

MTN Mobile Money Congo B. · Brazzaville, Republic of Congo

- Built a FastAPI-based customer support chatbot for MTN Mobile Money Congo covering 15 MoMo service topics using Azure OpenAI and PostgreSQL.
- Implemented a three-layer retrieval strategy: semantic search, keyword fallback, and full-context injection to reduce hallucinated responses.
- Collaborated with product, BI, engineering, and operations teams to validate topic coverage and improve answer reliability inside a large-scale fintech operation.

Deputy Head of Technology

Sep 2025 – Jun 2026

Enactus KNUST · Kumasi, Ghana

- Built internal tooling for mass email delivery, project data pipelines, and member management; contributed to chapter website and cross-division tech support.

PROJECTS

1. 10-Day AI/ML & Agentic Systems Sprint — nGOT Giants · KNUST, Apr–May 2026 | [GitHub](#)

12 hrs/day · 20+ projects across fintech, healthtech, agritech, and ecommerce

- **ETA Predictor** (Logistics) — ML model containerized with Docker, served via FastAPI; full experiment tracking via MLflow.
Stack: scikit-learn, FastAPI, Docker, MLflow, Pydantic V2, Poetry, pytest
- **Medical Literature RAG** (Healthtech) — Answers clinical questions with source citations; multi-step query decomposition over medical PDFs.
Stack: LlamaIndex, OpenAI GPT-4o-mini, text-embedding-3-small, Pinecone, FastAPI, Pydantic V2
- **Semantic Product Recommender** (eCommerce) — Natural-language queries ("cozy to wear in the rain") resolved via cosine similarity; returns top-N results with scores.

Stack: OpenAI text-embedding-3-small, Pinecone, FastAPI, Pydantic V2, Poetry

2. Lamla AI — Personalized AI-Powered Learning Platform | [GitHub](#) | [Live Demo](#)

Stack: Django, FastAPI, React, OpenAI GPT-4o-mini, Claude, PostgreSQL, RAG, Render, Vercel

- Built a dual-service backend: Django as API gateway with JWT auth and study plan management; async FastAPI microservice for low-latency LLM calls. Generates personalized quizzes, flashcards, and AI explanations from student-uploaded notes.

3. VeriMed — Medicine Authenticity Risk Assessment (Experimental) | [GitHub](#) | [Live Demo](#)

Stack: FastAPI, Next.js, EasyOCR, pyzbar, rapidfuzz, OpenCV, PostgreSQL

- Designed a multi-step verification pipeline: 3-image capture (packaging + barcode/QR), OCR extraction, registry fuzzy matching, and weighted risk scoring. Implemented an AI follow-up assistant with session-persisted conversation history tied to each verification result.

4. ScholarAid — Scholarship Management & Application Platform | [GitHub](#) | [Live Demo](#)

Stack: Django, DRF, Claude API, JWT, PostgreSQL, BeautifulSoup4

- Built a scraper pipeline that bulk-ingests scholarship portals via BeautifulSoup4; Claude API extracts structured fields (deadline, eligibility, amount) from raw URLs or pasted text. Students receive AI-powered essay feedback on submitted applications.

5. Tool-Augmented Chatbot — LLM Function-Calling DB Query | [GitHub](#)

Stack: FastAPI, SQLAlchemy, pyodbc, SQL Server, SQLite, basic RAG

- Built an LLM agent that routes queries across 3 strategies: calling 6 predefined SQL Server functions via function-calling, RAG over a document store, or direct answer — decided dynamically per query.

PROFESSIONAL TRAINING / CERTIFICATIONS

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| 1. 10-Day AI/ML & Agentic Systems Sprint | — nGOT Giants |
| 2. Working with the OpenAI API | — Data Camp |
| 3. LangChain for LLM Applications | — DeepLearning.AI |
| 4. ChatGPT Prompt Engineering for Developers | — DeepLearning.AI |
| 5. Introduction to FastAPI | — Data Camp |
| 6. Python Backend Development with Django | — FreeCode Camp |

SKILLS

Languages: Python (primary), Java, C++, SQL

Backend: FastAPI, Django, DRF, Flask, SQLAlchemy, PostgreSQL, REST APIs, Pydantic, Async

AI/LLMs: OpenAI API, Claude API, Azure OpenAI, Deepseek, Gemini, RAG, LangChain, LlamaIndex, Pinecone

Tools: Git, GitHub, Docker, Vercel, Render, Poetry, pytest, MS SQL Server, Claude Code, Cursor

LEADERSHIP & COMMUNITY

Deputy Head of Technology

Sep 2025 – Jun 2026

Enactus KNUST

International Relations Officer

Jan 2026 – Sep 2026

Science Students' Association (SCISA) — KNUST

VALUENETWORK — Community sharing opportunities and resources for African students and builders.

FoundersLens — TikTok page (40K+ followers) sharing insights from top founders and builders.